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Brain & Development 43 (2021) 843–850

BRAIN & DEVELOPMENT
Official Journal of the Japanese Society of Child Neurology

www.elsevier.com/locate/braindev

Original article

Risk of seizure recurrence after a first unprovoked seizure in childhood

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Received 13 January 2021; received in revised form 5 April 2021; accepted 20 April 2021

Abstract

Objective: The aim of this study was to assess the risk of recurrence after a first unprovoked seizure in childhood and to explore the correlation between the first and second seizures in recurrent patients.

Methods: In a prospective study, we included 467 children aged 1 month to 16 years, who were attended to between November 1, 2008 and October 31, 2016 following a first seizure. Children who had been started on treatment with antiepileptic drugs were excluded. Recurrence rates were calculated using Kaplan–Meier survival analyses. Univariate and multivariate analyses for recurrence risk were performed using the Cox proportional hazards model. The kappa coefficient of correlation for categorical data was calculated.

Results: Recurrences occurred in 280 children (60.0%), of which 75 (26.8%) occurred in the first month, 184 (65.7%) within 6 months, and 256 (91.4%) within 2 years. None of the patients had new neurologic sequelae after their first or second seizure. The estimates of seizure recurrence risk were 39.5%, 48.1%, 55.1%, 60.8%, 61.8% and 61.8% at 0.5, 1, 2, 5, 8, and 10 years after the first seizure, respectively. Multivariate analysis showed that abnormal electroencephalogram and neuroimaging findings significantly increased the risk of recurrence. First and second seizures were significantly associated with state of arousal, status epilepticus, and multiple seizures in recurrent patients.

Conclusion: Over half of untreated children had recurrence after a first unprovoked seizure, but prognosis was good overall.

Keywords: First unprovoked seizure; Children; Recurrence; Prognosis

1. Introduction

A practical clinical definition of epilepsy was published in 2014 [1]. Epilepsy is defined by any of the following conditions: (1) at least two unprovoked (or reflex) seizures occurring greater than 24 h apart; (2) one unprovoked (or reflex) seizure and a probability of further seizures, similar to the general recurrence risk (at least 60%) after two unprovoked seizures, occurring over the next 10 years; or (3) diagnosis of an epilepsy syndrome. While treatment with antiepileptic drugs prevents seizures, long-term medication can have side effects. According to the Japanese Epilepsy Society Guidelines [2], treatment initiation is recommended at the occurrence of a second seizure if informed consent is obtained from both the parents and child at the time of the first seizure. The estimate of recurrence risk from 16 studies was reported as 51% [3]. These included studies that involved children and adults, of which more than half were retrospective, half included treatment cases, and few were prospective studies that targeted a

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https://doi.org/10.1016/j.braindev.2021.04.005
0387-7604/

large number of untreated children. In a previous study, we reported the recurrence rate after a first unprovoked seizure in 250 untreated children [4]. The current study reports the long-term recurrence rate and risk factors after a first unprovoked seizure in a larger sample of Japanese children. We also report the correlation between the first and second seizures in recurrent patients, which was not reported in our previous study.

2. Materials and methods

We conducted a prospective observational study of patients who were attended to for their first unprovoked seizures at the Japanese Red Cross Society Himeji Hospital between November 1, 2008 and October 31, 2016. The study protocol and informed consent documents were approved by the institutional review board of the Japanese Red Cross Society Himeji Hospital. Written informed consent was obtained from parents prior to enrollment.

Patients were aged between 1 month and 16 years and were eligible if they had had a first unprovoked seizure and had received electroencephalogram (EEG) at our hospital. All possible children were examined by at least one of the authors. Children who had been diagnosed with definite epileptic seizure were included, while children who had had non-epileptic events and provoked seizures were excluded [5]. Children with repeated seizures within 24 h were included as multiple seizures. Children with a history of provoked seizures, such as febrile or neonatal seizures, were also included. Children with absence seizures, myoclonic seizures, epileptic spasms, and previous unprovoked seizures were excluded. Children who had started treatment with antiepileptic drugs after a first seizure were also excluded.

Status epilepticus was defined as more than 30 min of continuous seizure or two or more sequential seizures without full recovery of consciousness between seizures. Seizures were classified into focal seizures or generalized seizures; clinically apparent secondary generalized seizures or seizures with Todd's palsy were classified as focal seizures. A family history of unprovoked seizures was defined as seizures having occurred in first-degree relatives. The etiology was symptomatic if the child was known to have a static encephalopathy of pre- or perinatal origin, prior neurological insult, such as central nervous system infection, stroke, significant head trauma, mental retardation (estimated intelligence quotient below 70), or a motor disorder without any other defined etiology at first seizure [5]. Benign epilepsy with centrotemporal spikes and self-limited occipital epilepsies of childhood were classified as self-limited focal epilepsies (SLFE) [6]. Remaining children were classified as other, which was consistent with cryptogenic epilepsy [5].

EEGs were reviewed by at least one of the authors. Epileptiform abnormalities were classified into abnormal or no abnormality and non-epileptiform abnormalities were classified as normal. After enrollment, the children were followed-up at hospital visits or by telephone every 6 months to ascertain seizure recurrence. Recurrence was defined as an unprovoked seizure occurring more than 24 h after the first seizure. The follow-up period ended in December 2018.

Kaplan–Meier survival analysis was used to calculate the recurrence rates. The 95% confidence intervals (CIs) for the Kaplan–Meier estimates of recurrence risk were calculated at 6 months, 1, 2, 5, 8, and 10 years after the first seizure. Univariate and multivariate analyses were performed using the Cox proportional hazards model. The kappa coefficient of correlation for categorical data was calculated. A difference of $p < 0.05$ was considered significant. All statistical analyses were performed using JMP Version 14.2.0 (SAS Institute Inc., Cary, NC, USA).

3. Results

During the study period, 495 patients with a first unprovoked seizure visited the Japanese Red Cross Society Himeji Hospital. Fifteen patients were excluded because they either did not visit our hospital for EEG or were referred to their local hospital (i.e., resided outside of the area). Thirteen patients (six symptomatic, one SLFE, and six other) were excluded because they had received antiepileptic treatment after the first seizure. This resulted in a total of 467 patients to be included in the study. Table 1 shows the clinical characteristics of the patients. The mean age of the patients was 6.3 ± 4.4 years and the median age was 5.7 years. The mean follow-up period for patients who did not have recurrences was 5.6 years (range 0.03–10.1 years); 95.2% of patients were followed-up for greater than 2 years and 61.0% were followed-up for greater than 5 years. Thirty-three patients (7.1%) had abnormal neuroimaging findings on computer tomography (CT) or magnetic resonance imaging (MRI), and periventricular leukomalacia was the most common abnormality, which was observed in nine patients.

3.1. Overall recurrence risk

Of the 467 patients, 280 (60.0%) had recurrences during the study period. None of the patients had new neurologic sequelae after the first or second seizure. Seventy-five (26.8%) of the 280 recurrences occurred in the first month, 184 (65.7%) within 6 months and 256 (91.4%) within 2 years. Only two patients had recurrences more than 5 years after the first seizure. The Kaplan–Meier estimates of recurrence risk were 39.5% (95% CI, 35.1%–44.0%) at 6 months, 48.1% (95% CI,

Table 1
Clinical characteristics of patients.

	N	%
Sex		
Male	264	56.5
Female	203	43.5
Seizure type		
Focal	271	58.0
Generalized	196	42.0
Multiple seizures	61	13.1
Status epilepticus	43	9.2
Todd's palsy	15	3.2
State of arousal		
Awake	259	55.5
Sleep	194	41.5
Awake and sleep*	14	3.0
Etiology		
Symptomatic	82	17.6
SLFE	69	14.8
Other	316	67.7
Abnormal EEG	239	51.1
Neuroimaging abnormality on CT or MRI**	33	7.1
Prior febrile seizure	137	29.3
Family history of unprovoked seizure	49	10.5

SLFE, self-limited focal epilepsies; EEG, electroencephalogram; CT, computed tomography; MRI, magnetic resonance imaging. *Multiple seizure cases. **Four patients were not examined.

43.6%–52.6%) at 1 year, 55.1% (95% CI, 50.6%–59.6%) at 2 years, 60.8% (95% CI, 56.2%–65.3%) at 5 years, 61.8% (95% CI, 57.1%–66.3%) at 8 years, and 61.8% (95% CI, 57.1%–66.3%) at 10 years (Fig. 1). The estimates of recurrence risk at 5 years after being seizure free for 6 months, 1 year, and 2 years were 35.3% (95% CI, 29.7%–41.3%), 24.6% (95% CI, 19.2%–30.8%), and 12.7% (95% CI, 8.5%–18.7%), respectively. The estimates of recurrence risk at 10 years after being seizure free for 6 months, 1 year, 2 years, and 5 years were 36.8% (95% CI, 31.0%–43.1%), 26.4% (95% CI, 20.7%–33.0%), 14.8% (95% CI, 10.0%–21.5%), and 2.4% (95% CI, 0.6%–9.3%), respectively (Fig. 1).

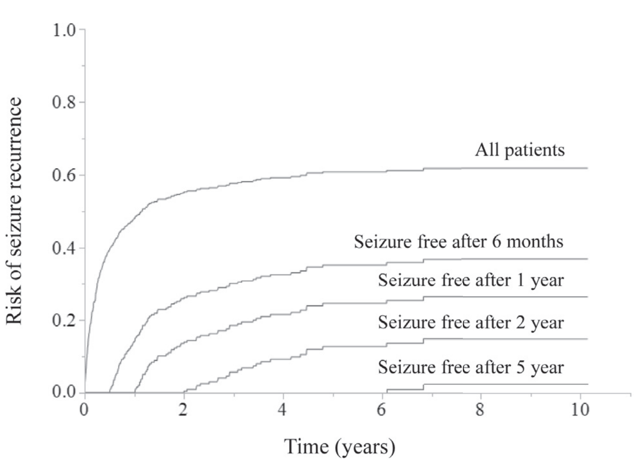

Fig. 1. Probability of seizure recurrence after a first unprovoked seizure and after various intervals without seizures. The Kaplan–Meier estimates of recurrence risk were 61.8% at 10 years. The graph shows the estimates of recurrence risk at 10 years after being seizure free for 6 months, 1 year, 2 years, and 5 years were 36.8%, 26.4%, 14.8%, and 2.4%, respectively.

3.2. Predictors of recurrence

Univariate analysis showed that the symptomatic and SLFE etiology groups had higher recurrence rates than the other etiology (Table 2). The Kaplan–Meier estimates of recurrence risk were 69.5% (95% CI, 58.8%–78.5%) at 2 years, 76.9% (95% CI, 65.3%–85.4%) at 5 years, and 76.9% (95% CI, 65.3%–85.4%) at 8 years in the symptomatic group. In the SLFE group, the estimates were 66.7% (95% CI, 54.8%–76.7%) at 2 years, 70.4% (95% CI, 58.4%–80.1%) at 5 years, and 70.4% (95% CI, 58.4%–80.1%) at 10 years. The other group had estimates of 48.9% (95% CI, 43.3%–54.3%) at 2 years, 54.7% (95% CI, 49.1%–60.3%) at 5 years, and 56.1% (95% CI, 50.3%–61.7%) at 10 years (Fig. 2). Additionally, univariate analysis showed a high recurrence rate for patients with epileptic EEG abnormalities, neuroimaging abnormalities on CT or MRI, and generalized seizures (Table 2). The Kaplan–Meier estimates of recurrence risk were 64.4% (95% CI, 58.2%–70.2%) at 2 years, 70.5% (95% CI, 64.2%–76.1%) at 5 years, and 71.6% (95% CI, 65.1%–77.3%) at 10 years in the abnormal EEG group, compared with 45.2% (95% CI, 38.8%–51.7%) at 2 years, 50.5% (95% CI, 43.9%–57.1%) at 5 years, and 51.4% (95% CI, 44.6%–58.1%) at 8 years in the normal EEG group (Fig. 3). The Kaplan-Meier estimates of recurrence risk were 78.8% (95% CI, 61.7%–89.5%) at 2 years, 94.3% (95% CI, 73.2%–99.0%) at 5 years in the abnormal neuroimaging group, compared with 53.6% (95% CI, 48.8%–58.2%) at 2 years, 58.6% (95% CI, 53.7%–63.2%) at 5 years, and 59.6% (95% CI, 54.7%–64.4%) at 10 years in the normal neuroimaging group (Fig. 4).

Full model multivariate analysis was conducted with 11 variables (Table 2). The recurrence rate of the abnormal EEG group remained significantly higher than the normal EEG group. The recurrence rate of the abnormal neuroimaging group also remained significantly higher than the normal neuroimaging group. The recurrence rates of the symptomatic etiology, SLFE etiology, and generalized seizures groups were no longer significant.

3.3. The correlation between the first and second seizures

The correlations of state of arousal, status epilepticus, and multiple seizures between the first and second seizures were evaluated in 280 recurrent patients (Table 3). Out of 144 patients who had the first seizure during awakening, 124 (86.1%) had the second seizure during awakening. Ninety-five (77.9%) of 122 patients

Table 2
Effect estimates for the univariate and multivariate models.

	Univariate analysis			Multivariate analysis		
	Rate ratio	95% CI	*p*	Rate ratio	95% CI	*p*
Etiology						
Symptomatic/Other	1.7	1.3–2.3	0.0003	1.2	0.8–1.8	0.32
SLFE/Other	1.4	1.0–1.9	0.04	1.1	0.7–1.6	0.76
Symptomatic/SLFE	1.2	0.6–1.2	0.28	1.2	0.7–1.9	0.58
Abnormal EEG	1.7	1.3–2.1	<0.0001	1.6	1.2–2.1	0.001
Neuroimaging abnormality on CT or MRI	2.2	1.5–3.2	<0.0001	1.8	1.1–3.1	0.02
Generalized seizure	1.3	1.1–1.7	0.02	1.3	1.0–1.6	0.08
Seizure while asleep	1.2	1.0–1.6	0.10	1.1	0.8–1.4	0.70
Male	1.0	0.8–1.3	0.88	0.9	0.7–1.2	0.65
Multiple seizures	1.0	0.7–1.4	0.98	1.1	0.8–1.7	0.53
Status epilepticus	0.9	0.6–1.4	0.79	1.0	0.6–1.6	0.99
Todd's palsy	1.1	0.6–2.0	0.79	1.0	0.5–2.0	0.92
Prior febrile seizure	1.0	0.8–1.3	0.84	1.1	0.8–1.4	0.71
Family history of unprovoked seizure	1.2	0.8–1.7	0.34	1.2	0.8–1.7	0.35

CI, confidence interval; SLFE, self-limited focal epilepsies; EEG, electroencephalogram; CT, computed tomography; MRI, magnetic resonance imaging.

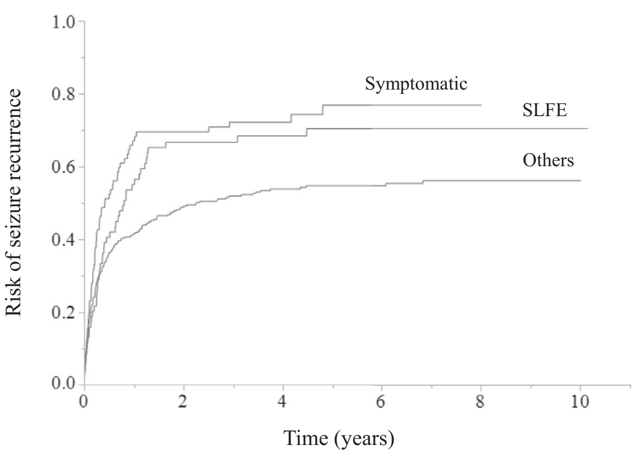

Fig. 2. Probability of seizure recurrence after a first unprovoked seizure as a function of etiology. Univariate analysis showed that the symptomatic and SLFE etiology group had higher recurrence rates than the other etiology group, but this difference was not significant in the multivariate analysis. SLFE: self-limited focal epilepsies.

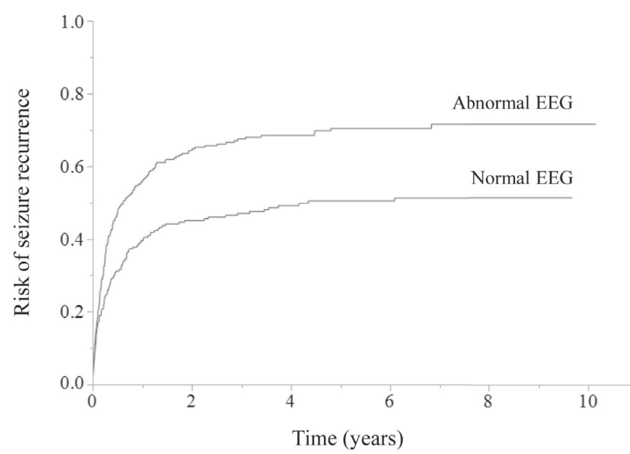

Fig. 3. Probability of seizure recurrence after a first unprovoked seizure as a function of the electroencephalogram. Univariate and multivariate analyses showed the abnormal EEG group had higher recurrence rates than the normal EEG group. EEG: electroencephalogram.

who had the first seizure during sleeping, had the second seizure during sleeping. State of arousal of the first and second seizures were significantly consistent (Kappa coefficient 0.64, $p < .0001$). Nine (33.3%) of 27 patients who had status epilepticus at the first seizure had status epilepticus at the second seizure, while six (2.4%) of 251 patients whose first seizure was not status epilepticus had status epilepticus at the second seizure (Kappa coefficient 0.39, $p < .0001$). Six (17.6%) of 34 patients who had multiple seizures during the first seizure had multiple seizures at the second seizure, while 12 (4.9%) of 246 patients whose first seizure did not involve multiple seizures had multiple seizures at the second seizure (Kappa coefficient 0.16, $p = .002$). The correlations between the first and second seizures by etiology is shown in Table 3.

4. Discussion

This large-scale prospective study investigated the recurrence rate in untreated children after a first unprovoked seizure, with a long-term follow-up. There have been many studies investigating seizure recurrence rates after a first unprovoked seizure. However, many of these studies have small sample sizes, include adults and treated patients, or are retrospective studies [3]. Prospective studies of large numbers of children have reported recurrence rates at 2 years after a first unprovoked seizure ranging between 33% and 57% [7–11]. Furthermore, Shinnar et al. and Ramos Lizana et al. reported recurrence rates of 42% and 64%, respectively, at 5 years after a first unprovoked seizure, though 14% and 16% of

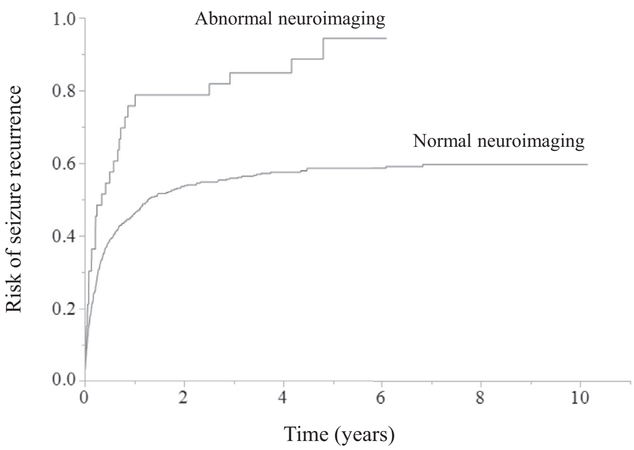

Fig. 4. Probability of seizure recurrence after a first unprovoked seizure as a function of neuroimaging. Univariate and multivariate analyses showed that the abnormal neuroimaging group had higher recurrence rate than the normal neuroimaging group.

patients of the respective study samples had been treated [7,8]. In our study, we showed similar recurrence rates of 55.1% at 2 years and 61.8% at 5 years. In contrast to most previous reports, we also reported the probability of seizure recurrence after various intervals without seizure recurrence. Mizorogi et al. reported that the recurrence rate of children who were seizure free for 2 years was 11.8%, at 4 years after the initial seizure [12], which is similar to our results. About two-thirds of recurrent patients had recurrence within 6 months, although the recurrence rate after 6 months of being seizure free was still moderately high at 35.3% at 5 years and 36.9% at 10 years after the initial seizure.

We revealed that abnormal EEG, symptomatic and SLFE etiologies, abnormal neuroimaging findings, and generalized seizures were found to be predictors of recurrence in the univariate analysis, whereas EEG abnormalities and neuroimaging abnormalities were found to be predictors of recurrence in the multivariate analysis. Abnormal EEG has been reported most frequently as a risk factor for recurrence [7,9,11–17]. Our results demonstrated that EEG is important for predicting recurrence. There have been reports that patients with a symptomatic etiology have significantly higher recurrence rates than idiopathic/cryptogenic etiologies [7–9,16,17]. In our study, we did not use the classifications of idiopathic or cryptogenic etiology, often used in previous studies [7–9], because most of our idiopathic patients were classified as SLFE and patients with idiopathic generalized epilepsy with absence and myoclonic seizures were excluded. Furthermore, diagnosis of epilepsy with generalized tonic-clonic seizures alone (EGTCA), which is defined as idiopathic generalized epilepsy, is difficult at first seizure. Moreover, although the other etiology group ultimately included cryptogenic etiology and EGTCA, only three patients were suspected to have EGTCA at the first seizure. Shinnar et al. and Stroink et al. reported that when patients with idiopathic and cryptogenic etiologies were grouped together, symptomatic etiology patients had a higher recurrence risk than idiopathic/cryptogenic etiology patients [7,9]. Ramos et al. reported that the recurrence risk was highest for symptomatic, followed by idiopathic, with cryptogenic etiology carrying the lowest risk; there were significant differences between all groups [8]. Neuroimaging abnormalities were also shown to be a predictor of recurrence in several reports [8,18,19]. Stroink et al. showed that abnormal brain CT was associated with a higher recurrence rate in a univariate, but not in a multivariate, analysis [8]. In contrast, neuroimaging abnormalities were not investigated as a predictor of recurrence in the reports by Shinnar et al. and Ramos et al. [7,9]. In the present study, abnormal neuroimaging findings were associated with higher recurrence rates in the multivariate analysis, whereas etiology was not. Specifically, recurrence rate was low in patients with the symptomatic etiology without neuroimaging abnormalities. Therefore, our results suggest that neuroimaging plays an important role in predicting the risk of recurrence. The symptomatic group with normal neuroimaging included patients with motor paralysis, in addition to intellectual disability and autism spectrum disorder. Some studies have reported that patients with focal seizures had a significantly higher recurrence rate than those with generalized seizures, though multivariate analyses did not reveal any significant differences in any of these studies [7,10,12,13]. We showed that patients with generalized seizures had a significantly higher recurrence rate than those with focal seizures in the univariate analysis, but this was not observed in the multivariate analysis. Because most seizures were classified based on interviews with family members or teachers, it is possible that some of the focal or secondary generalized seizures are considered generalized seizures. It would be difficult to assess recurrence rate of true seizure classifications; results are likely to vary depending on the study.

To our knowledge, no study has investigated the correlation between the first and second seizures. We found that state of arousal of the first and second seizures was significantly consistent. Patients with status epilepticus or multiple seizures during the first seizure had a significantly higher risk of status epilepticus or multiple seizures, respectively, in the second seizure. Furthermore, status epilepticus was significantly different for all etiologies of symptomatic, SLFE, and other. State of arousal of the first and second seizures was significantly different in the symptomatic and other etiology but not in SLFE. Forty-one of 47 patients in the SLFE group had first and second seizures during sleep. Patients with SLFE may not have significantly differed because they were more likely to have seizures during sleep. These results

Table 3
The correlations between the first and second seizures by etiology.

a) All

	Second seizure			Second seizure			Second seizure	
First seizure	Awake	Sleep	First seizure	Status epilepticus	No status epilepticus	First seizure	Multiple seizures	Single seizure
Awake	124	20	Status epilepticus	9	18	Multiple seizures	6	28
Sleep	27	95	No status epilepticus	6	245	Single seizure	12	234
Kappa coefficient 0.64, $p < .0001$			Kappa coefficient 0.39, $p < .0001$			Kappa coefficient 0.16, $p = .002$		

b) Symptomatic

	Second seizure			Second seizure			Second seizure	
First seizure	Awake	Sleep	First seizure	Status epilepticus	No status epilepticus	First seizure	Multiple seizures	Single seizure
Awake	26	7	Status epilepticus	4	6	Multiple seizures	1	7
Sleep	5	20	No status epilepticus	2	49	Single seizure	5	48
Kappa coefficient 0.58, $p < .0001$			Kappa coefficient 0.42, $p = .0002$			Kappa coefficient 0.03, $p = .39$		

c) SLFE

	Second seizure			Second seizure			Second seizure	
First seizure	Awake	Sleep	First seizure	Status epilepticus	No status epilepticus	First seizure	Multiple seizures	Single seizure
Awake	1	2	Status epilepticus	2	1	Multiple seizures	0	2
Sleep	3	41	No status epilepticus	1	44	Single seizure	2	44
Kappa coefficient 0.23, $p = .056$			Kappa coefficient 0.64, $p < .0001$			Kappa coefficient −0.04, $p = .38$		

d) Other

	Second seizure			Second seizure			Second seizure	
First seizure	Awake	Sleep	First seizure	Status epilepticus	No status epilepticus	First seizure	Multiple seizures	Single seizure
Awake	97	11	Status epilepticus	3	11	Multiple seizures	5	19
Sleep	19	34	No status epilepticus	3	152	Single seizure	5	142
Kappa coefficient 0.56, $p < .0001$			Kappa coefficient 0.26, $p < .0001$			Kappa coefficient 0.23, $p = .0004$		

may be useful for predicting state of arousal and the risk of status epilepticus and multiple seizures when a second seizure occurs.

Several studies have consistently shown that treatment of a first unprovoked seizure decreases the risk of recurrence but does not affect the probability of long-term remission in both children and adults [20–22]. In fact, the effects associated with drugs are not uncommon and may include serious life-threatening adverse events and chronic effects that affect quality of life [20]. Therefore, in Japan, the recommended approach in children is to not treat following the first unprovoked seizure and begin treatment if a second seizure should occur [2]. Patients who have a high risk of recurrence or are prone to status epilepticus in the second seizure may be considered for treatment after the first seizure. However, the absence of patients with neurological sequelae from a total of 747 first and second seizures (even if they had status epilepticus), supports the approach of no treatment after the first seizure. Treatment with antiepileptic drugs may be considered in circumstances where the benefits of reducing the risk of a second seizure outweigh the risks of pharmacologic and psychosocial adverse events [23].

Our study has two limitations. First, there may have been a patient bias for seizure status and etiology because of the hospital setting. However, our hospital covers the population of Naka-Harima and Nishi-Harima, west of Hyogo, Japan, and provides pediatric emergency care and specialized outpatient care in pediatric neurology; many children with possible epilepsy visit our hospital because of an emergency or referral from regional clinics. Therefore, the present cohort is likely to be similar to the general population. Second, 13 children were excluded because they received antiepileptic treatment after the first seizure. However, this amounted to 3% of the total sample and would therefore not significantly affect the results.

In conclusion, we estimated that 61.8% of Japanese children who had had a first unprovoked seizure, had a second seizure within the following 10 years. Abnormal EEG and neuroimaging findings were predictors of recurrence. First and second seizures were significantly associated with state of arousal, status epilepticus, and multiple seizures in recurrent patients.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

We thank Sarina Iwabuchi, PhD, from Edanz Group (https://en-author-services.edanz.com/) for editing a draft of this manuscript.

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